

Hemalata Nayak

College of William and Mary, Virginia, USA | hnayak@wm.edu | +1-757-332-0512 | [Linkedin](#) | [GitHub](#)

EDUCATION

College of William & Mary, PhD in Data Science Jan 2024 – Ongoing

- **Coursework:** Bayesian Reasoning, Network Analysis, Linear algebra & Numerical Methods, Deep Learning, Generative AI and Trustworthy AI.

Central University of Karnataka, M.Sc in Physics Nov 2021 – Aug 2023

- GPA: 8.4/10
- **Coursework:** Classical Mechanics-I & II, Quantum Mechanics - I & II, Nuclear and Particle Physics, Statistical Mechanics, Electronics, Spintronics, Computational Physics, Solid States Physics
- **Lab courses:** Condensed Matter Physics, Nuclear Physics, Electronics, General Physics

Berhampur University, B.Sc in Physics Aug 2017 – Sep 2020

- GPA: 8.49/10

RESEARCH PROJECTS

National Institute of Education and Research, Bhubaneswar, India Sep 2023 – Dec 2023

- **Project Title: Characterization of Silicon Photomultiplier (SiPM)**
Silicon Photomultipliers (SiPMs) are cutting-edge semiconductor devices designed for precise and efficient detection of low-intensity light signals. These detectors are vital components in various fields such as medical imaging, high-energy physics, and remote sensing. In this work, I conducted characterization of SiPMs to ensure reliable and accurate performance. Through rigorous characterization, one can determine important parameters like the photon detection efficiency, dark count rate, and time resolution. This information helps in optimizing SiPMs for specific applications, ensuring they meet the performance requirements, and minimizing any potential sources of error or noise.

Master Thesis Feb 2023 – June 2023

- **Project Title: GNN-based energy reconstruction for INO-ICAL**
One of the important parameters in neutrino oscillation experiments is the energy of the neutrino through which the mixing parameters are also determined. In this context, while a basic Kalman filter based algorithm is currently used for energy reconstruction from the detector hits, I developed a Graph Neural Network based approach to verify its efficiency and speed compared to the conventional technique. This work was presented in an International conference and the conference proceeding is out now. ([MSc Thesis](#))

SKILLS

- **Physics toolkit:** ROOT, Geant4
- **Programming Languages:** Python, C++
- **Machine Learning Frameworks:** PyTorch, Scikit-learn
- **Tools & Libraries:** NumPy, Pandas, Matplotlib, Git, Linux

PUBLICATIONS

- Presented poster at the 16th International Conference on Heavy Quarks and Leptons (HqL 2023), TIFR Mumbai ([certificate](#))
- **GNN-based muon energy reconstruction for INO-ICAL**
Hemalata Nayak, Deepak Samuel, Lakshmi P. Murgod (Karnatak University)
Published in *PoS HQL2023 (2024), 075*. ([Proceeding](#))
- **Physics Event Classification Using Large Language Models**
Cristiano Fanelli, James Giroux, Patrick Moran, **Hemalata Nayak**, Karthik Suresh, Eric Walter.

[\(Paper\)](#)

- **AI-Assisted Detector Design for the EIC (AID(2)E)**

M. Diefenthaler, C. Fanelli, L.O. Gerlach, W. Guan, T. Horn, A. Jentsch, M. Lin, K. Nagai, **H. Nayak**, C. Pecar, K. Suresh.

[\(Paper\)](#)

PROFESSIONAL SERVICE

- Served as Event Manager for the AI4Fusion Summer School 2024. [\(website\)](#)
- Served as Event Manager for the AI4Fusion Summer School 2025 [\(website\)](#)
- AI4EIC paper spotlight committee. [Website](#)

CERTIFICATION

- Completed an intensive Parallel Computing Pathway Program, organized by the National Center for Supercomputing Applications (NCSA) and the Pittsburgh Supercomputing Center (PSC), featuring lectures and hands-on assignments. [\(Certificate\)](#)